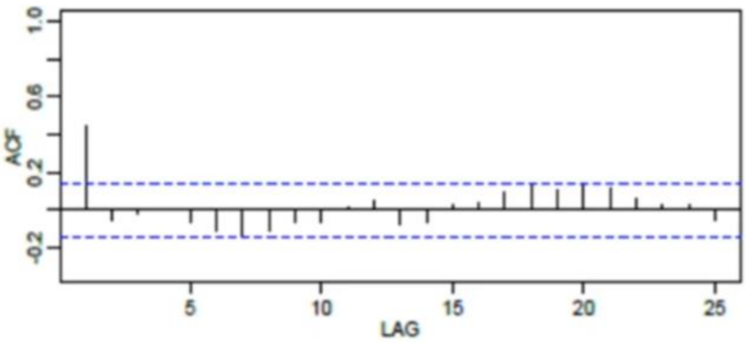
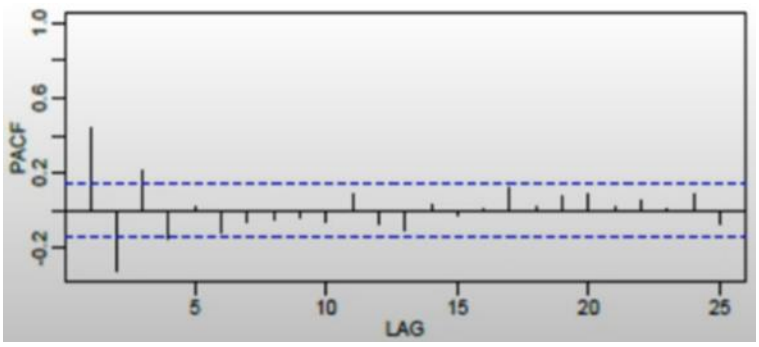
General Identification Rules Using ACF & PACF

Term	Meaning	Visual Behavior
Cut Off	Becomes insignificant after a certain lag	Significant spikes up to lag k, then zero
Tail Off	Decreases gradually over many lags	Slowly declining spikes

Model Type	ACF Behavior	PACF Behavior	Interpretation Rule
AR(p)	Tails off gradually (exponential or damped sine wave)	Cuts off sharply after lag *p*	If PACF cuts off at lag *p*, choose AR(p)
MA(q)	Cuts off sharply after lag *q*	Tails off gradually	If ACF cuts off at lag *q*, choose MA(q)
ARMA(p,q)	Tails off gradually	Tails off gradually	If both tail off, consider ARMA
White Noise	No significant spikes	No significant spikes	No AR or MA structure

Q.2.





Step 2: Look at the ACF Plot (First Image)

What do we observe?

- There is **one significant spike at lag 1**
- After lag 1, all values are inside the confidence bounds (blue dotted lines)
- No gradual decay pattern
- No repeated significant spikes

This means the **ACF cuts off after lag 1**
This is the signature behavior of an **MA(1) model**.
If it were:

- MA(2) → significant spikes at lag 1 and lag 2
- MA(3) → significant spikes up to lag 3
- AR model → ACF would decay gradually, not cut off sharply

Step 3: Look at the PACF Plot (Second Image)
What do we observe?

- PACF does **not show a clean cutoff**
- Several small spikes appear
- It gradually fluctuates within bounds

That is exactly what we expect for an MA model:
PACF tails off instead of cutting off sharply

Step 4: Why NOT AR(1)?
If this were AR(1):

- PACF would show one clear significant spike at lag 1 and then cut off
- ACF would show gradual decay

But here:

- ACF cuts off
- PACF tails off

So AR(1) is not consistent with the pattern.

Step 5: Why NOT ARMA?
In ARMA:

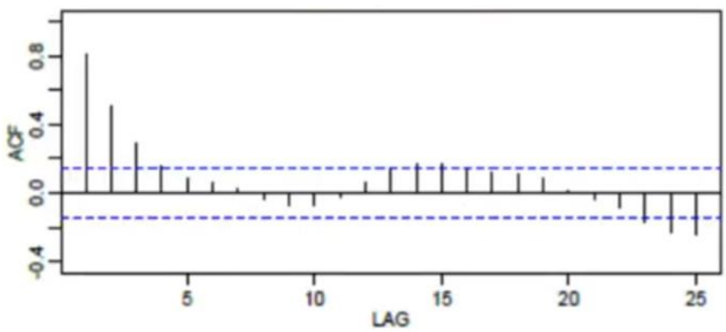
- Both ACF and PACF tail off gradually
- No sharp cutoff

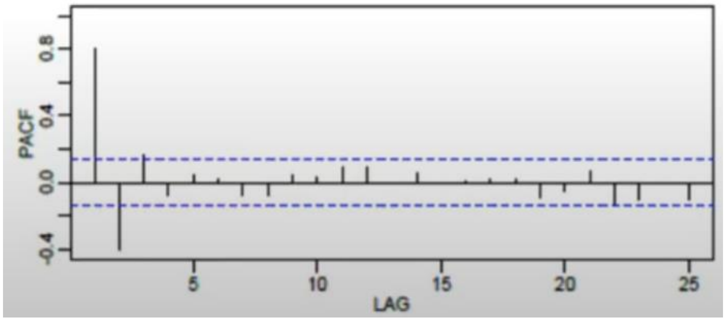
But here:

- ACF clearly cuts off at lag 1

So ARMA is unlikely.

Q.3.





Answer: ARMA (1,1)

Step 1: Observe the ACF (Left Plot)

What we see:

- Large spike at lag 1
- Gradually decreasing values afterward
- No sharp cutoff
- Slow decay pattern

Interpretation:

- The ACF is **tailing off**, not cutting off.
- So this is **NOT a pure MA(q)** model (because MA should show ACF cutoff).

Step 2: Observe the PACF (Right Plot)

What we see:

- Significant spike at lag 1
- Possibly lag 2
- Then gradual decay
- No sharp cutoff at a specific lag

Interpretation:

- PACF is also **tailing off**
- So this is **NOT a pure AR(p)** model (because AR should show PACF cutoff).

Step 3: Why Not Pure AR?

For AR(p):

- PACF must cut off sharply at lag p.

Here:

- PACF does not clearly stop at lag 1 or 2.
- It slowly decays.

Therefore → Not pure AR.

Step 4: Why Not Pure MA?

For MA(q):

- ACF must cut off sharply at lag q.

Here:

- ACF gradually decreases.
- No clear cutoff at lag 1 or 2.

Therefore → Not pure MA.